SKY LIS · NATIONAL TECHNOLOGY

Development Status and Proposed Changes

Part A records what the Sky LIS application is today — modules, pages, platform foundations, verification and the installed state. Part B sets the Client Portal prototype v3 and the Super Admin prototype v4.1 against that application, item by item, and lists the decisions and candidate changes they imply.

DOCUMENT
LIS-DS-001 · v1.2

DATE
2026-10-20 (v1.1: 2026-10-19, v1.0: 2026-10-18)

SOFTWARE STATE
v0.14.1 (github.com/arfaawadnt/Sky-LIS); 0.14.0 installed on WIN-9S2KO4FA64L

PROTOTYPES COMPARED
Client Portal v3 · Super Admin v4.1 (supplied 2026-10-18)

SOURCES
README release history, docs/deploy/Installations.md, docs/build/MODULES.md, docs/srs (LIS-SRS-001 v1.1), docs/validation, docs/knowledge/sky-lis/SkyLIS_Prototypes_v3-v4.1_Difference.md

RELATED
LIS-RP-001 Remaining points, LIS-BP-001 Build plan, LIS-VR-001 / LIS-VR-002 Validation reports, LIS-UAT-001

AUDIENCE
Business owner and project management, National Technology

STATUS
Prepared by the development team for review

1. Executive summary

The Release-1 scope of Sky LIS — the pages of the v2.1 client and v3.1 admin prototype builds — is built, validated twice, and installed on the development server of National Technology. Four change requests of 13–14 October were delivered and installed within the same two days. **Update 2026-10-20 (v1.2):** after the first use of 0.15.2 the business owner reversed D-2: *Payments* returns to the menu and the treasury is removed from the system (change request LIS-CR-004; CP-28, CP-29 and BR-27 withdrawn; payments carry cashier and branch only). **Update 2026-10-19 (v1.1):** the business owner asked for the *Not built* / *Partial* items of the comparison; they were built the same day as change request LIS-CR-003 (ten items: A1, A2, C3, C9, C10, C11, C18, C19, C21, C22 — see sections 5 and 11), verified by every gate and in the browser, and await release v0.15.0 and installation. The new prototypes (client v3, admin v4.1) are supersets of the prototypes Release 1 was specified from: of their 36 proposals, 21 exist as proposed, 6 exist in part, 4 exist in another form and 5 are not built. None of the 5 requires a change of the data model; the largest is a user-interface consolidation.

45
Release-1 pages delivered (28 client · 17 admin) of 89 specified in the Business SRS

15
business and platform modules built (Release-2 backends included)

2,722
automated tests green after LIS-CR-003 (1,661 backend · 1,061 frontend; 2,488 at v0.14.0)

19
installations on WIN-9S2KO4FA64L, 0.3.2 → 0.14.0, each with backup and smoke test

36
prototype proposals compared: 31 built · 3 partial · 2 different · 0 not built (before LIS-CR-003: 21 · 6 · 4 · 5)

1
business decision still open (D-1 terminology); D-2, D-3 and D-4 were settled by the request of 2026-10-19

Built Partial Different Not built Release 2 / parked

PART A

Current development

What has been built, how it was verified, and where it runs — as of release v0.14.1 (2026-10-14) plus change request LIS-CR-003 of 2026-10-19, built and verified in the repository and awaiting its release (v0.15.0) and installation.

2. Product shape and architecture

Sky LIS is a cloud, multi-tenant laboratory information system with two portals on one backend: the **Client Portal** used by each laboratory (reception, laboratory, finance, set-up, account) and the **Super Admin Portal** used by National Technology (clients, subscriptions, master data, platform). A laboratory subscribes on the public site, verifies the manager's mobile and receives a fully provisioned tenant with the global catalogue copied in.

Backend

- .NET 9 / C# 13, Clean Architecture in six projects (`Platform.*`), CQRS with MediatR, FluentValidation, EF Core 9 on PostgreSQL 17.
- Tenant isolation by composite keys `(tenant_id, id)` and PostgreSQL row-level security on every tenant table; the runtime role cannot bypass it.
- 27 API controllers, about 500 operations under `/api/v1` ; 96 tables in 15 migrations; one idempotent deployment script.
- Every command and query carries a permission policy; architecture tests enforce the layer rules, the policy rule, atomic transitions and the migration script freshness.

Frontend

- Angular 22 (zoneless, signals), strict TypeScript 6, Vitest; 79 pages on a shared UI kit (tables with server-side paging and sorting, drawers, dialogs, tabs, print documents).
- English and Arabic (RTL) on every page; language and results density saved as user preferences.
- Release scope switch (`RELEASE_SCOPE = 'R1'`): Release-2 pages exist behind it and are hidden from menu and routes.
- Session rules proven in validation: one identity per tab, cross-tab sign-out, idle timeout with warning, branch of the session.

3. Modules and Release-1 pages

Each module was specified in an intake document (`docs/intake/LIS-INT-<MOD>-001`), built in a wave with its own build slot and database, and integrated through documented event contracts. The SRS numbers the pages CP-nn (client) and SA-nn (admin).

MODULE	DELIVERED IN RELEASE 1	BUILT, PARKED FOR RELEASE 2	STATE
Foundation	Sign-in, refresh, branch of the session, password recovery (OTP), audit ledgers, outbox, notifications, tenant settings, idle timeout	MFA / TOTP, step-up re-authentication, retention purge	Built
Account & onboarding	Public site, Subscribe wizard (OTP shown on the page until an SMS gateway exists), Login, Recovery, Branch selection, Dashboard (CP-05), Profile, laboratory logo	Wallet view, Support tickets, client audit log page	Built
Organization	Branches, Roles & privileges (permissions, special rights, result rights per clinical group × branch), Users with overrides, Parameters	—	Built
Catalogue A / B	Clinical groups, Services (members, print alone, header), Tests (types, reference ranges by age and gender, list values with colour, formula, paragraphs, parent services), Service sequence, Sample types, Units, Preparations, Comment library, Doctors, Microbiology (drug types, commercial names, antibiotics, organisms, sensitivity values)	Patient conditions, Additional services, Diagnoses, Sending labs, Shortcut sets, Test rules, Reasons, Report templates (client pages; all are Release-1 pages on the admin portal)	Built
Pricing	Payers with contract tiers, Base price lists (per-service prices, effective dates, CSV), price lookup for registration, Price query (CP-12)	Contract pricing page (factors, per-service overrides, exclusions, alerts — backend complete)	Built
Patients	Patient register and file (demographics, Egyptian national-ID fill, documents, merge, similar-name check), VIP masking, identity lock by result status	—	Built

MODULE	DELIVERED IN RELEASE 1	BUILT, PARKED FOR RELEASE 2	STATE
Registration	New registration desk (CP-06: patient + visit, service entry with typeahead, prices with patient / payer share, discount, preparations, payment on the page, labels, job order), Search registrations (CP-10) with row actions, visit edit, Price query	Home reservations, batches, transfers, cancellations register	Built
Samples	Auto-collect / auto-receive on save (decision A-03); samples dialog from the visit row	Collection, Samples by visit, Rejections, Tracking pages	Partial by decision
Results	Worklist (CP-18), Results by accession (CP-19: entry, review → verify with delta acknowledgement, amend, undo, comments, print, previous results, hidden-by-rights notes, previous / next visit)	Results by sample, Work queue, Quality control, analyser landing	Built
Billing & Treasury	Payments register (CP-23) with refunds, Treasuries, Treasury access (device-bound requests, SoD with the administrator exception), collections report	Batch payment, Cash closing	Built
Reporting	Work amount report (eight groupings, CSV, report-days limit), dashboard KPIs	Claims, the sixteen financial reports	Partial by scope
Platform administration	Platform dashboard, Clients (source, status, reset password), Subscriptions & wallets (plans, hourly charge run), Sessions, Audit log, Countries, Governorates, Platform parameters, Notification templates, Admin roles and users	Integration library (analyser models, PCR kits, code systems, rule templates), Support desk	Built
Integration & Automation	—	Analysers, analyser tests and samples, site agents, PCR kits, verification settings, medical rules (backends and result intake built)	Release 2

Business SRS pages: 45 of 89 are Release 1 (green) and delivered; the remaining 44 are Release 2 by the scope decision of LIS-RP-001, most of them with a finished backend.

4. Platform foundations delivered

Security

- Credential envelope (the browser seals passwords and codes with a per-request key), Argon2id hashing, ES256 access tokens of 15 minutes, encrypted subject identifiers.
- Refresh-token rotation with reuse detection and family revocation, strict cookie scope, origin and CSRF checks, rate-limit matrix per endpoint.
- Password policy and forced change, mobile numbers in E.164 and unique in every spelling, idle timeout per tenant, one identity per browser tab.
- Start-up guards refuse a non-LIS database, foreign configuration keys and relaxed security in production.

Audit and operations

- Field-level change log, hash-chained audit trail per tenant with a verify-chain endpoint, security-event ledger; all append-only at database level.
- Transactional outbox with a hosted processor, lease, back-off and dead letter; domain policies react to events (visit registered, sample collected, patients merged …).
- Notifications over SMS gateway, SMTP or a pickup folder; templates editable on the admin portal with built-in fallback.
- Windows service and IIS site installers, verified backups, restore drills, smoke test, runbook and installation records.

5. Change requests delivered 13–14 October 2026

Every request was specified against its intake, built with tests, released with notes and installed on the server the same day. Versions are those of the README release history.

VERSION	DATE	CHANGE	INSTALLED
`v0.9.0`	2026-10-13	**Change request of 13 October (four items).** Three new special rights (demographics editable up to a result status, treasury report of the whole branch, report time limit in days) on roles and per-user overrides; patient demographics editable from Search registrations with the identity lock by result status (`PAT-032`); treasury (collections) report on Payments by method, cashier or branch; the new **Work amount report** page under Finance with eight groupings, print and CSV; Results by accession criteria fold into a chip bar after a query.	2026-10-13

VERSION	DATE	CHANGE	INSTALLED
`v0.10.0`	2026-10-13	**Results by accession redesigned.** One sticky context strip replaces the header, the criteria card and the matching list once a visit is open; criteria in a dialog, matching visits in a side panel, unit tabs, sample chips and action buttons on one row. The first result row moves from about 470 px to about 110 px below the top bar; a single match opens at once with the cursor on the first editable cell.	2026-10-14
`v0.11.0`	2026-10-13	**Results by accession follow-ups.** Previous / next visit from the strip (and Ctrl+Alt+← / →) across the pages of the list or by accession number; hints folded behind the *i* button; *Compact rows* saved as a user preference (migration `MembershipResultsDensity`); the strip stays pinned on narrow screens.	2026-10-14
`v0.12.0`	2026-10-14	**Catalogue and rights.** Parent services on the test page (join or leave profiles and cultures under the service rules); Results by accession names the results a unit hides by result rights; new clinical groups are granted automatically to the roles that manage roles; review stack for browser checks beside the installed service.	2026-10-14
`v0.13.0`	2026-10-14	**Subscription wizard shows the verification code** (interim, until an SMS gateway is connected), controlled by the host setting `Lis:Notifications:RevealCodes`; the start-up guard refuses it once an SMS gateway is configured.	2026-10-14
`v0.14.0`	2026-10-14	**Treasury access: the laboratory's primary administrator approves their own request** (named exception of SOD-TRS-001). Everyone else keeps the segregation of duties; nobody can reject their own request. The session profile carries `isPrimaryAdministrator`.	2026-10-14
`v0.14.1`	2026-10-14	Installation record of 0.14.0 (documentation only).	n/a

VERSION	DATE	CHANGE	INSTALLED
`v0.15.0`	2026-10-19	**Change request LIS-CR-003 — prototype v3 / v4.1 gaps** (`docs/build/CR-2026-10-19_Prototype-Gaps.md` , migration `PrototypeGaps`). Super Admin: client login logs (A1) and the location hierarchy governorate → city → district → area (A2). Client: second phone and location pick lists on the patient (C3), the Accession details page (C9), the menu change — treasuries, treasury access and payments reached from the pages that need them (C11), payment methods as set-up with a per-method reference rule (C10), service sequence reordering by drag / arrows / typed position inside a clinical-group filter (C18), payer contacts (C19), price list Excel download / upload, bulk adjustment and print (C21), the service prices page (C22). Six packages built in parallel build slots and integrated; gates: 106 architecture, 883 domain, 367 application, 149 integration, 156 functional, 1,061 frontend specs, lint, boundaries, production build; browser check of every new page on the review stack.	Release pending

6. Verification and quality

Automated gates (every release)

SUITE	TESTS	COVERS
Domain unit tests	795	Aggregates, state transitions, codes, segregation of duties
Architecture tests	106	Layer rules, policy per request, atomic transitions, migration script, banned APIs
Application unit tests	327	Handlers, validators, crypto primitives, configuration guard
Integration tests (PostgreSQL)	142	Row-level security, interceptors, audit chain, outbox, migrations up and down

Validation campaigns and acceptance material

- **LIS-VR-001** (2026-10-11): first full validation — suites, every Release-1 page of both portals, public pages and the backend cycle from screen to database; defects fixed and re-tested, released as v0.4.0.
- **LIS-VR-002** (v0.5.0, 2026-10-11/12): second full validation with the verification of every fix — 32 defects (30 distinct) fixed with a test and re-tested in the browser and the database; 64 earlier items re-checked; released as v0.6.0.
- **LIS-UAT-001**: Release-1 acceptance scripts per persona (44 critical scripts dry-run, three rounds), with negative tests and the Arabic check.
- **Backend cycle** (`tests/validation/backend-cycle.mjs`): provisions a laboratory and drives registration → samples → results → payment against the deployed API with database assertions; run after every installation that touches the backend.
- Each installation: clean build from the tagged worktree, dry-run of both installers, verified backup, migration check, smoke test by host name and by IP (20 / 20), rollback folders kept.

SUITE	TESTS	COVERS
WebApi functional tests	145	End-to-end flows per module through the HTTP edge and the database
Frontend specs (Vitest)	973	Pages, facades, interceptors, session rules, UI kit
TOTAL	**2,488**	**PLUS ESLINT, BOUNDARY CHECK AND PRODUCTION BUILD**

7. Installed state

WIN-9S2KO4FA64L — production lane

INSTALLED BUILD	**0.14.0** (2026-10-14); repository tip v0.14.1
API	Windows service `NTLIS`, `http://127.0.0.1:5100`, virtual service account
SITE	IIS `lis-ui`, `https://WIN-9S2KO4FA64L` and `https://192.168.80.38:5443`, self-signed certificate
DATABASE	PostgreSQL 17, private instance on 5433 as service `LIS_POSTGRES`; 15 of 15 migrations applied
BACKUPS	daily scheduled task 02:00 to `E:\lis-backups`, verified dumps before every upgrade
NOTIFICATIONS	pickup folder only; `RevealCodes = true` while no SMS gateway is connected
DATA	demo laboratory and validation data; primary administrator flagged for the demo laboratory

Installation timeline

`2026-10-10`	0.3.2 first installation; 0.3.5 the same day
`2026-10-11`	0.4.0 (validation fixes, first migration applied), 0.5.0 (registration desk), 0.5.1
`2026-10-12`	0.5.2, 0.5.3, 0.6.0 (second validation fixes), 0.7.0 / 0.7.1 (laboratory brand), 0.7.2 (parameters page)
`2026-10-13`	0.8.0 (idle timeout, report templates, notification delivery), 0.9.0 (change request of 13 October)
`2026-10-14`	0.10.0, 0.11.0 (results page), 0.12.0, 0.13.0 / 0.13.1 (code on the subscribe page), 0.14.0 (treasury self-approval)

Nineteen installations in five days, each recorded in `docs/deploy/Installations.md` with build hash, backup, downtime window and smoke-test result. Typical API downtime per upgrade: about 20 seconds.

Not installed: the test lane (`NTLIS_TEST`, `lis-ui-test`), an SMS / e-mail gateway, a trusted certificate and public host name, off-machine backup copies. These are the pilot prerequisites of LIS-RP-001 (Phase B).

8. Open items carried from Release 1

Business decisions pending

- Demographics edit limit, branch collections and report-days defaults for existing roles (Organization intake A14–A16).
- Collections report scope and grouping defaults (Billing intake Q20–Q22); work amount report questions (Reporting intake Q12–Q18).
- Treasury self-approval exception confirmed for the primary administrator only (Treasury intake §9) — implemented as decided on 2026-10-14.
- VIP masking rule for users without *See VIP* (Patients Q8 / Registration Q12) — implemented as an assumption.
- Phase-A decisions of LIS-RP-001 still open: public host name and certificate, master-data update policy, number formats, who prints.

Engineering items parked

- SMS gateway connection (then switch `RevealCodes` off in the same configuration change).
- MFA / TOTP and step-up re-authentication; retention purge and other scheduled jobs beyond the wallet charge.
- OpenAPI document and generated TypeScript client.
- Release-2 pages: samples, contract pricing, bookings, batches, cash closing, claims and financial reports, quality control, integration and automation screens.
- Data migration from LDM, load test and penetration test (LIS-RP-001 Phase B).

PART B

Proposed changes — prototypes v3 / v4.1

What the new prototypes propose beyond the ones Release 1 was built from, what the application already does, and what is left to decide and build.

9. Method and scope of the comparison

The two files supplied on 2026-10-18 are **supersets of the earlier prototypes**: the v3 client keeps everything v1 had and adds layers labelled v2, v2.1 and v3 (patches on the navigation, the dictionary and the pages); the v4.1 admin keeps the v2 screens and reworks the catalogue. The comparison therefore treats the *delta* as the proposal. Each prototype was inventoried structurally (navigation, pages, fields, buttons, grids, validation texts, dictionary patches), diffed against its predecessor, and every resulting proposal was checked against the built code — domain model, API contracts, routes and pages — not against documents.

26
client-portal proposals (C1–C26)

10
admin-portal proposals (A1–A10)

0
proposals that need a change of the stored data model

10. What the prototypes propose

10.1 Client Portal v3

AREA	PROPOSAL (DELTA AGAINST V1)
Terminology	*Service* → **Test**; *Services / Packages* → **Profiles / Panels** (a profile contains tests, a panel contains profiles); *Payer* → **Contract**; *Contract* → **Relationship** (price list + patient share %); *Query service* → *Query test*.
Navigation	Added: Dashboard (insights), Accession details, Payment methods, Test price. **Removed:** Treasuries, Request treasury, Treasury requests, Search payments — payments handled inside registration / accession details.
Dashboard / Insights	Filters by registration date, contract and branch; KPIs (registrations, payments, active ordered items, unique patients, wallet); charts: daily registrations and payments, top-20 doctors and contracts, share / gender / paid-remaining / test-mix pies.

AREA	PROPOSAL (DELTA AGAINST V1)
Registration	Patient block with second phone, e-mail, ID, insurance ID, address; contract → relationship selection; catalogue picker with LOINC search; patient and contract share per item; discount on the patient share; **several payments inside registration** with method and transaction number; payment summary with refund due; payment history; preparation answers on the page.
Accession details (new)	One page per accession: visit information, ordered tests, cancel a test with reason, collect sample with confirmation, enter a result, previous results, open payments.
Set-up	Payment methods (code, name, requires transaction no.); test configuration v2.1 (ranges by age, optional values with colour, formula check, culture set-up, aliases, *exists in profiles*); profiles / panels editor; global sequence by drag, arrows or typed position; contracts with a contact block (phone 1 / 2, WhatsApp, e-mail) and relationships; price lists with **Excel template / upload**, **bulk price adjustment** with rounding and print; Test price page; preparations with a default answer; grids with "in use → set inactive".

10.2 Super Admin Portal v4.1

AREA	PROPOSAL (DELTA AGAINST V2)
Navigation	Adds **Client login logs**; *Services* → **Profile / Panel**; global sequence of tests; *Export Excel* → *Export CSV*.
Client login logs	Per client user session: login / logout time (UTC), IP, computer / mobile, OS and browser, duration, status; filters and CSV.
Location set-up	Governorate → city → district → area, edited inside the governorate.
Countries	Flag as an uploaded image (JPG / PNG / SVG) beside dial code, currency, time zone and mobile length.
Dashboard	Operational period, KPIs, active / inactive clients donut, registrations by top laboratories, recent clients.
Catalogue	Tests configured as on the client v2.1; profile / panel editor with ordered members and type rules; drug types as dosage forms; commercial names carry a drug type.
Roles, users, clients	Screen privileges per role; admin users with country code and mobile; clients with source, reset password and activation; the main Super Admin protected.

11. Difference against the developed application

Status: Built exists as proposed · Partial part of it exists · Different exists with another model, name or place · Not built absent.

11.1 Client portal (C1–C26)

#	PROTOTYPE V3 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
C1	Terminology: Test / Profile / Panel, Contract / Relationship	Service types Test, Profile, Package, Culture; a profile holds tests, a package holds services (rules CAT-021/022/027/036). Payer → contract tiers → contract pricing per base price list with patient / payer factors and per-service shares.	Different	Same structures, different names: Package ≈ Panel, Payer ≈ Contract, Contract tier ≈ Relationship. A dictionary and SRS change; the data model needs nothing.
C2	Dashboard / Insights with period, contract and branch filters, top-20 lists, pies, test mix	CP-05 dashboard (registrations, collected today, samples received, pending / unverified / delayed, rejected, wallet, 7-day chart, recent activity, quick actions); Work amount report by day, payer, contract tier, user, doctor, clinical group, service or branch with CSV.	Partial	The numbers exist; the filterable insights page with ranked lists and pies is not built.
C3	Registration: second phone, e-mail, ID, insurance ID, address on the patient block	Patient: names, gender, DOB / age, national ID, nationality, condition, mobile, e-mail, governorate, city, address, VIP; visit: insurance ID, treatment no.	Built	Delivered 2026-10-19 (LIS-CR-003): second phone on the patient (file, register, picker, similar check, CSV); location pick lists when the governorate has cities. Insurance ID stays on the visit (as in LDM).
C4	Contract & relationship selection; patient / contract share per item	CP-06: payer / contract chooser, service entry with typeahead, line prices with patient and payer share from the contract pricing.	Built	
C5	Discount as value or percentage on the patient share only	Visit discount as amount or percent, cap from the special right, reason; applied to the patient share.	Built	

#	PROTOTYPE V3 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
C6	Several payments inside registration, method + transaction number when required, refund due, payment history	Registration takes one payment on the page (treasury approved for the device); further and partial payments and refunds on Payments (CP-23) by accession, with a reference field; refunds with reasons.	Different	The money flow is complete but split across Registration and Payments; the prototype folds it into the visit. Treasury / device approval (BR-27) has no counterpart in v3.
C7	Result date = max test duration, no. of tests	Expected result per line from the test's duration and minimum duration; line count on the page.	Built	
C8	Continue to registration from Query test	CP-12 price query → registration.	Built	
C9	Accession details page: visit info, ordered tests, cancel with reason, collect sample, enter a result, previous results, payments	Search registrations row actions: edit, patient, samples, payments, documents, results, cancel services with reason; Results by accession (entry, review, verify, print, previous results); sample pages exist in the backend and the Release-2 menu.	Built	Delivered 2026-10-19 (LIS-CR-003): one Accession details page (search strip, header with the patient drawer, ordered tests with cancel, samples with label / collect, results status, payments and refunds, documents). Auto-collect stays on in Release 1, so *Collect* rarely shows (D-3).
C10	Payment methods as set-up (code, name, requires transaction no.)	Fixed methods Cash, Card, POS, Transfer with an optional reference on every payment.	Built	Delivered 2026-10-19 (LIS-CR-003): tenant payment methods (code, names, kind, requires reference, order), seeded with the four defaults; the cash desk and the registration payment panel use them.

#	PROTOTYPE V3 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
C11	Treasuries, treasury requests and Search payments removed from the menu	Treasuries, Treasury access (device approval, SoD with the administrator exception), Payments register with the collections report.	Built	Superseded on 2026-10-20 by LIS-CR-004: the treasury pages and requests are removed from the system; *Payments* (with the collections report) is in the menu.
C12	Test types Numeric / Non-numeric / Optional list / Formatted text / Formula / Culture	Test types Numeric, Non-numeric, Optional list, Restricted list; formula expression on the test; culture through the Culture service type with organisms, antibiotics and sensitivity values; formatted text as paragraphs on results.	Built	Naming only (Restricted list ≈ optional list without free text).
C13	Reference ranges by age (D/M/Y, inclusive), gender, normal and critical bounds, non-numeric comments	Range bands with age unit days / months / years, gender restriction, low / high, critical low / high, range resolver; reference text printed per range.	Built	
C14	Optional list values with colour, normal / critical flags, default	List values with value, normal, critical, default, placeholder and colour.	Built	
C15	Test short name, alias names, *Exists in profiles*	Code, short name, LOINC, CPT, methodology; parent services on the test page (2026-10-14); aliases absent.	Partial	Alias names (search synonyms) are the only gap.
C16	Duration (hours) and working days for the result date	Duration hours, minimum duration hours, test days (working days).	Built	
C17	Profiles / Panels editor: members, print alone, header, alias, LOINC unique	Services editor: members (tests / services) with sequence, print alone, report header, LOINC; type rules CAT-021/022/036.	Built	Panel ≈ Package.

#	PROTOTYPE V3 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
C18	Global sequence: drag / arrows / typed position, clinical-group filter	CP-42 service sequence: a sequence number typed per service, saved per row.	Built	Delivered 2026-10-19 (LIS-CR-003): drag and drop, ↑ / ↓, Alt+↑ / Alt+↓, typed position, clinical-group filter; the visible rows keep their own set of numbers.
C19	Contracts: contact details (phone 1 / 2, WhatsApp, e-mail)	Payer: name, search key, web portal user, default flag, status; contract tiers; no contact block.	Built	Delivered 2026-10-19 (LIS-CR-003): phone 1 / 2, WhatsApp and e-mail on the payer, with CSV export.
C20	Relationships with price list and patient share %	Contract pricing: priority, base price list, patient factor, payer factor, per-service overrides, exclusions, alert condition.	Built	Richer than the proposal.
C21	Price lists: Excel template download / upload with validation, bulk adjustment with rounding, print	Base price lists with per-service prices, effective dates, CSV export.	Built	Delivered 2026-10-19 (LIS-CR-003): Excel download / validated all-or-nothing upload, bulk adjustment (amount / percent, rounding), printable list.
C22	Test price page (which price lists price a test)	Price query by visit (CP-12); price list detail per list.	Built	Delivered 2026-10-19 (LIS-CR-003): Finance › Service prices — one service across the base price lists (editable inline) and the contract tiers.
C23	Preparations: question types, ordered options with a default, assigned tests	Preparations with kinds Text, Yes / No, Option list, gender restriction, options, service assignment; answers captured at registration; no default option.	Partial	Small: a default answer per question.

#	PROTOTYPE V3 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
C24	Patient drawer: edit, visit history, open results / payments	Patient file (CP-11) with visits, documents, merge; patient drawer from the visits register (2026-10-13).	Built	
C25	Set-up grids: pagination, "in use → set inactive"	Master-data pages: server-side paging and sorting, deactivate instead of delete, read-only drawer without the edit right.	Built	
C26	Subscription wizard, OTP, branch choice, forgot password	CP-02 / CP-03 / CP-04 built; the OTP is shown on the page until an SMS gateway exists.	Built	

11.2 Super Admin portal (A1–A10)

#	PROTOTYPE V4.1 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
A1	Client login logs: login / logout times, IP, device, OS / browser, duration, filters, CSV	SA-08 Sessions: live sessions per client user with last seen, revoked at, IP address and user agent, revoke action; SA-09 audit log and the security-event ledger (sign-in, sign-out, refusals).	Built	Delivered 2026-10-19 (LIS-CR-003): history grid per session family with device / OS / browser parsed from the user agent, duration, status, filters and CSV.
A2	Location set-up: governorate → city → district → area	Countries and governorates (with the mobile length per country); patients carry governorate, city (free text) and address.	Built	Delivered 2026-10-19 (LIS-CR-003): cities, districts and areas edited from the Governorates page; pick lists on the patient form (D-4 settled by the request). The client address stays free text.
A3	Countries: dial code, currency, time zone, mobile length, flag image	Country: dial code, currency and code, language, time zone, mobile length, flag as text.	Built	Flag as an uploaded image instead of a code is a small change.

#	PROTOTYPE V4.1 PROPOSAL	DEVELOPED APPLICATION (V0.14.1)	STATUS	NOTE
A4	Dashboard with operational period, KPIs, clients donut, registrations by top labs, recent clients	SA-03 platform dashboard: period, client counts, admin users, activity totals, wallets at or below zero, plan distribution, registrations by client, recent clients.	Built	
A5	Roles with screen privileges; main Super Admin protected	SA-06 roles with catalogue permissions and special rights; primary administrator protected (TEN-012).	Built	
A6	Admin users with country code + mobile; privileges from the role only	SA-07: mobile in E.164 (unique), role, no overrides in the platform tenant.	Built	
A7	Clients grid with source, reset password, activation	SA-04 clients: source Subscription / Manual, status, reset client password, activate / deactivate, wallets and plans (SA-05).	Built	
A8	Profile / Panel catalogue, tests with configuration, global sequence of tests, aliases	SA-12 master data mirrors the client catalogue (services, tests with ranges / lists / formula, sequence, sample types, units, preparations, comments), copied to new laboratories.	Built	Aliases absent (as C15); naming as C1.
A9	Drug types as dosage forms; commercial names with a drug type	SA-14 drug types, commercial names, antibiotics, organisms, sensitivity values.	Built	
A10	Export CSV on every grid	CSV export on every register (UTF-8 BOM, formula-safe).	Built	

11.3 Totals (after LIS-CR-003, 2026-10-19)

PORTAL	BUILT	PARTIAL	DIFFERENT	NOT BUILT	TOTAL
Client portal	21	3	2	0	26
Super Admin portal	10	0	0	0	10
BOTH	**31**	**3**	**2**	**0**	**36**



12. Built, but absent from the prototypes

The developed application covers the whole Release-1 scope of the Business SRS and a number of later decisions that the v3 / v4.1 prototypes do not show. Adopting the prototypes literally would remove them; the recommendation is to keep every item below.

- **Treasury** (CP-28 / CP-29): treasuries per branch, device-bound access requests, approval with segregation of duties and the primary-administrator exception, revocation; cash accepted only on an approved treasury (BR-27).
- **Payments register and refunds** (CP-23), the **collections report** by method / cashier / branch with the branch right; cash closing (Release-2 backend).
- **Work amount report** with eight groupings, CSV and the per-user report-days limit.
- **Special rights** (discount cap, pay, refund, transfer, cancel, modify verified, preview unverified, print invoice, see VIP, demographics edit limit, branch collections, report days) and **result rights** per clinical group × branch.
- **Patient identity lock** by result status and VIP masking.

- **Results by accession** as built: sticky context strip, criteria dialog, matching-visits panel, previous / next visit, review → verify with delta acknowledgement, amend with reason, undo, comments and internal notes, print lists, report templates with logo and signature, hidden-by-rights notes.
- **Audit**: field changes, hash-chained audit trail, security events, verify-chain endpoint, audit log pages.
- **Platform**: subscriptions and wallets (plans, charge run), notification templates, platform parameters, sessions, 26 tenant settings including time zone and session timeout.
- **Security**: credential envelope, refresh-token rotation with reuse detection, idle timeout, password policy, mobile uniqueness, rate limits, row-level security per tenant.
- **Catalogue extras and Release-2 backends**: patient conditions, reasons, diagnoses, additional services, report templates, test rules, shortcut sets, sending labs, doctors; analysers, site agents, PCR kits, verification settings, medical rules.

13. Decisions for the business owner

Four decisions preceded the work on the proposals. D-2, D-3 and D-4 were settled on 2026-10-19 by the request to build C9 / C11, C9 and A2 (recorded in the rows below); D-1 stays open.

ID	DECISION	QUESTION	RECOMMENDATION
D-1	Terminology	Adopt the v3 words — Test / Profile / Panel, Contract / Relationship, Global sequence, Query test — in both portals? Dictionary, SRS and intake wording change only; the stored model stays.	Decide once for both portals. LDM-trained users know *Service / Package / Payer*; a rename costs little but must be complete (screens, documents, training).

ID	DECISION	QUESTION	RECOMMENDATION
D-2	Payments inside the visit vs. the Payments page	The prototype removes *Search payments* and the treasury pages and folds payments into registration / accession details.	Keep the Payments register and the treasury controls (BR-27 and the self-approval decided on 2026-10-14). Optionally show the visit's payment history on the registration edit page. **Settled 2026-10-19:** the controls stay; the pages left the sidebar and are reached from the accession page and the pages that need them (C11). **Reversed 2026-10-20 (LIS-CR-004):** Payments is back in the menu; the treasury, its pages, device approval and BR-27 are removed; money movements carry cashier, branch, method and reference.
D-3	Sample collection in the client portal	The prototype confirms collection per visit; Release 1 runs with auto-collect / auto-receive on (decision A-03).	Keep as decided (Release 2) unless the pilot laboratory needs manual collection. **Settled 2026-10-19:** auto-collect stays on; the accession page offers *Collect* only for a sample still Ordered (C9).
D-4	Location hierarchy	City → district → area as new master data with pick lists, or keep free-text city and address?	Pick lists only if the pilot's reporting needs them; otherwise keep free text and revisit with the claims module. **Settled 2026-10-19:** pick lists on the patient when the governorate has cities, free text otherwise; the client address stays free text (A2).

14. Candidate changes by size

Sizes are relative to the delivered change requests (a *small* item is of the order of the treasury self-approval of v0.14.0; a *medium* item of the order of the work amount report of v0.9.0). Durations are indicative engineering effort including tests, release and installation, and assume the decisions above.

SIZE	CHANGE	REF.	INDICATIVE EFFORT	DEPENDS ON
Small	**Delivered 2026-10-19 (LIS-CR-003).** Payment methods per tenant with *requires reference* (or a mandatory reference for Card / POS / Transfer)	C10	1–2 days	—
Small	**Delivered 2026-10-19 (LIS-CR-003).** Contact block on the payer (phone 1 / 2, WhatsApp, e-mail)	C19	1 day	D-1 (naming)

SIZE	CHANGE	REF.	INDICATIVE EFFORT	DEPENDS ON
Small	Alias names on tests (search synonyms, used by the service entry typeahead)	C15, A8	1–2 days	—
Small	**Delivered 2026-10-19 (LIS-CR-003) for C3; A3 (flag image) stays open.** Second phone on the patient; flag image on the country	C3, A3	1–2 days	—
Small	**Delivered 2026-10-19 (LIS-CR-003).** Test price view: one service across every price list, with edit	C22	1–2 days	—
Small	Default answer per preparation question	C23	½ day	—
Small	Terminology rename of both portals and the documents (if D-1 says yes)	C1, A8	2–3 days	D-1
Medium	**Delivered 2026-10-19 (LIS-CR-003).** Price list Excel import (validated, row-level messages) and bulk adjustment with rounding; printable price list	C21	1–2 weeks	—
Medium	Insights dashboard: period / contract / branch filters, top-20 doctors and contracts, share / gender / paid-remaining / test-mix charts (data already served by the work amount report)	C2	1–2 weeks	—
Medium	**Delivered 2026-10-19 (LIS-CR-003).** Client login history grid (device from the user agent, duration, filters, CSV) over the stored sessions and security events	A1	1 week	—
Medium	**Delivered 2026-10-19 (LIS-CR-003).** Location hierarchy master data and address pick lists on patient and client	A2	1–2 weeks	D-4
Medium	**Delivered 2026-10-19 (LIS-CR-003).** Global sequence: drag-and-drop and move up / down, reordering within a clinical-group filter	C18	1 week	—
Medium	Payment history and further payments on the registration edit page (keeping the Payments register)	C6	1 week	D-2

SIZE	CHANGE	REF.	INDICATIVE EFFORT	DEPENDS ON
Large	**Delivered 2026-10-19 (LIS-CR-003).** Accession details as one page (visit, lines, cancel, collection, results, payments) — mostly a consolidation of existing pages; sample collection is a scope decision	C9	2–3 weeks	D-2, D-3

> **Not recommended:** removing Treasuries, Treasury access and Search payments from the client menu (C11). They implement BR-27 and the decision of 2026-10-14, and the collections report depends on the Payments register.

15. Suggested way forward

STEP	CONTENT	OUTCOME
1	Business owner takes decisions D-1 … D-4 (one review session with this document and the difference analysis).	Scope of the change set fixed
2	Small items as one change request (C10, C19, C15, C3 / A3, C22, C23, and the rename if decided), released and installed together.	About one week; one release
3	Medium items in priority order chosen by the business — suggested: Excel import and bulk adjustment (C21), insights dashboard (C2), login history (A1), then sequence reordering (C18), payment history on the visit (C6), location hierarchy (A2) if D-4 says yes.	Each its own release, 1–2 weeks apart
4	Accession details page (C9) after D-2 and D-3, or deferred to Release 2 with the sample pages.	Decision-dependent
✓	**Done 2026-10-19:** steps 2–4 were delivered together as LIS-CR-003 for A1, A2, C3, C9, C10, C11, C18, C19, C21 and C22 (release v0.15.0 pending); still open from the plan: C2 insights dashboard, C6 payment history on the visit, C15, C23, A3 and the rename (D-1).	One release
–	In parallel and independent of the prototypes: SMS gateway (then `RevealCodes` off), trusted certificate and host name, test lane, the Phase-A decisions of LIS-RP-001.	Pilot prerequisites

LIS-DS-001 v1.2 · 2026-10-20 · Prepared by the development team of Sky LIS for National Technology. Facts in Part A come from the repository (release history, installation records, module map, test suites); facts in Part B from the structural comparison of the

supplied prototypes against the built code. Every status in section 11 was checked against the domain model, contracts, routes and pages of v0.14.1. Efforts in section 14 are indicative and exclude business review time.